Diagnosing Learned Embeddings for Duplicate Track Removal in Line Segment Tracking

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Abstract

Line Segment Tracking (LST) reconstructs charged-particle trajectories for the CMS Phase-2 High Level Trigger, and removes duplicate track candidates using a learned embedding trained with a contrastive loss. We do not propose a replacement for that method; we diagnose it. Working from the same 500-event $t\bar{t}$ sample and the same feature definitions, we report four measurements that the existing description does not contain. First, the embedding is strongly rank-deficient: its effective rank is 3.96 of 12 dimensions, with 98.2% of the variance in three principal components, which accounts for the previously unexplained observation that increasing the embedding dimension yields no improvement in test loss. Second, a globally learned Mahalanobis precision matrix on top of such an embedding is a reparameterisation of the Euclidean metric rather than a richer model, so the common Euclidean-versus-Mahalanobis comparison is degenerate by construction; we verify this numerically and give the conditions under which it fails. Third, the $\Delta R^2 < 0.02$ window used to form candidate pairs is a blocking step with a measurable recall ceiling of 0.9952 for T5–T5 pairs and 0.9634 for pLS–T5 pairs, which bounds every downstream recall figure. Fourth, balanced pairwise AUC substantially overstates deployment performance: an AUC of 0.9566 corresponds to an end-to-end duplicate recall of 0.188 at a realistic duplicate prior and a 99% track-efficiency requirement. We additionally show that duplicate decisions taken pairwise are globally inconsistent, and that the choice of which duplicate to discard — currently a fixed collection priority — is learnable.

1 Introduction

Line Segment Tracking [1, 2] builds track candidates hierarchically from mini-doublets and triplets into quintuplets (T5), and combines these with pixel line segments (pLS) from an upstream pixel iteration. Because the same simulated particle can be reconstructed by several candidates, and because candidates from different collections may describe the same trajectory, the final candidate list must be cleaned of duplicates. In production this is done by "cross-cleaning": candidates close in η–ϕ are compared, and a fixed collection priority (pT5 > pT3 > T5 > pLS) decides which survives.

A learned component was added to this step [3]: two networks embed T5 (30 features) and pLS (10 features) candidates into a shared 6-dimensional space, and the Euclidean distance between embeddings enters a contrastive loss [4] that pulls duplicates together and pushes non-duplicates apart. The method is deployed, and reduces the duplicate rate by up to 50% for displaced tracks at a throughput cost of 3–4%.

This paper is diagnostic. We take that method as given and ask what its behaviour actually is, using measurements that are standard elsewhere in machine learning but have not been applied here. Our contributions are:

1. **The embedding is rank-deficient** (Section 3). Its effective rank [6] is roughly half its nominal dimension. This explains, mechanistically, why raising the embedding dimension does not help — an observation reported in [3] without explanation.

2. **The Euclidean-versus-Mahalanobis comparison is degenerate** (Section 4). We show, and verify numerically, that a globally learned precision matrix on top of an encoder ending in a trainable linear layer adds no expressive power. We give the three conditions under which this ceases to hold, and a per-instance alternative that escapes it.

3. **The pairing window has a measurable recall ceiling** (Section 5), and **balanced AUC overstates deployment performance** (Section 6).

4. **Pairwise decisions are globally inconsistent** (Section 7), and **arbitration is learnable** (Section 8). We also calibrate the decision threshold with a distribution-free guarantee on efficiency loss (Section 9).

Relation to prior work. We claim neither the learned-embedding approach [3] nor learned duplicate arbitration in tracking, which has been demonstrated in ACTS using a clustering step followed by a ranking network [16, 15]. The equivalence in Section 4 is a known result for linear metric learning [8, 9, 10]; our contribution is to observe that it survives a deep encoder and therefore applies to the deployed configuration, and to quantify its consequences. Related recent work introduces pair-conditioned Mahalanobis metrics in a different domain [11], and the "gauge freedom" framing for neural representations [12].

2 Setup

Data. A sample of 500 simulated $t\bar{t}$ events at HL-LHC pileup. T5 candidates are described by the 30 features of [3]: per-anchor-hit η, ϕ, z, r differences along the candidate, inverse inner/bridge/outer radii, and the T3 fake/prompt/displaced scores with their differences. pLS candidates use 10 features including η, its fitted error, ϕ, inverse p_T, and circle-fit quantities.

Pairs. Candidate pairs are formed within $\Delta R^2 < 0.02$, matching the production cross-cleaning window. A pair is labelled *duplicate* if both members match the same simulated track index. We generate 1,000,000 T5–T5 pairs. Pair sampling is capped per event, which we show in Section 7 has consequences for graph-level measurements.

Training. Each metric is trained with its *own* pair of encoders under its *own* objective. This matters: sharing one encoder across metrics, or reading a differently-trained embedding out with a new distance, produces numbers that reflect the training objective rather than the metric. Encoders are $30 \to 128 \to 64 \to d$ with batch normalisation and no terminal nonlinearity. The margin is set from the scale the embedding actually occupies rather than fixed a priori. This is a live failure mode in both directions: a margin far above the embedding scale leaves the repulsive term permanently unsaturated, while a margin far below it clamps that term to zero and removes the repulsive gradient entirely, after which the embedding collapses. We therefore log the fraction of non-duplicate pairs still inside the margin each epoch and treat values near 0 or 1 as a failed run; all results reported here held this fraction between 0.29 and 0.56.

3 The embedding is rank-deficient

We measure the effective rank of the embedding using RankMe [6], the exponentiated Shannon entropy of the normalised singular-value spectrum,

$$\mathrm{RankMe}(Z) = \exp\left(-\sum_k p_k \log p_k\right), \qquad p_k = \frac{\sigma_k(Z)}{\|\sigma(Z)\|_1} + \epsilon, \tag{1}$$

which equals the nominal dimension for an isotropic embedding and 1 for a fully collapsed one. Table 1 reports it for each metric.

The Euclidean embedding attains an effective rank of 3.96 out of 12, with 98.2% of the variance in three components. This is dimensional collapse in the sense of [7]: under a contrastive objective the gradient on each singular direction is proportional to that direction's own singular value, so directions that begin small remain small.

Consequence: the dimension scan measures collapse, not sufficiency. Ref. [3] reports that "increasing the embedding dimension up to 32 shows no improvement in test loss for either model", based on a scan over embedding sizes with five random seeds. That is usually read as evidence that six dimensions suffice. We tested the alternative reading directly by sweeping $d \in \{4, 6, 8, 12, 16, 32\}$ (Figure 1, Table 4).

The Euclidean embedding's effective rank is essentially independent of d: over an eight-fold increase in nominal dimension it rises only from 3.47 to 4.44, a factor of 1.28, while balanced AUC is flat to the fourth decimal (0.9551 at $d = 4$, 0.9554 at $d = 32$). We therefore reproduce the reported null result and supply its mechanism: the additional dimensions are never populated, so they cannot contribute. At the deployed $d = 6$ the embedding uses an effective 3.59 of its six dimensions.

The same sweep shows this is a property of the metric and not of the task. Under a cosine objective the effective rank tracks the nominal dimension over the same range, rising from 3.93 to 20.78 — a factor of 5.28 against the 1.28 of the Euclidean case — at equal or better discrimination. Six dimensions are not "enough"; six dimensions are what the Euclidean objective can fill.

Replication. Trained independently on two disjoint halves of the events, the effective rank agrees to 4.87 versus 4.82 (Euclidean) and 10.26 versus 10.49 (cosine), with balanced AUC agreeing to 0.003. The separation between the two metrics is an order of magnitude larger than the spread between halves. This controls for subset and seed instability only; it shares whatever biases the sample carries and is not a test of external validity.

The metric choice determines whether collapse happens. Table 1 shows the effect is not uniform across metrics. The cosine embedding retains an effective rank of 11.05 of 12 where the Euclidean one retains 3.96, and it does so at no cost in discrimination: cosine matches Euclidean on T5–T5 pairs and is substantially better on the cross-collection pLS–T5 pairs (0.9493 versus 0.9004). Cosine is scale-invariant, so the direction along which the Euclidean embedding expands — and which comes to dominate its spectrum — simply does not exist for it. On this evidence, a cosine objective is the better default for this task.

A variance floor does not prevent rank collapse. It is tempting to prevent collapse with VICReg's variance term [5], which lower-bounds each dimension's standard deviation. We find this insufficient: with every per-dimension variance held inside its bounds, the effective rank still falls

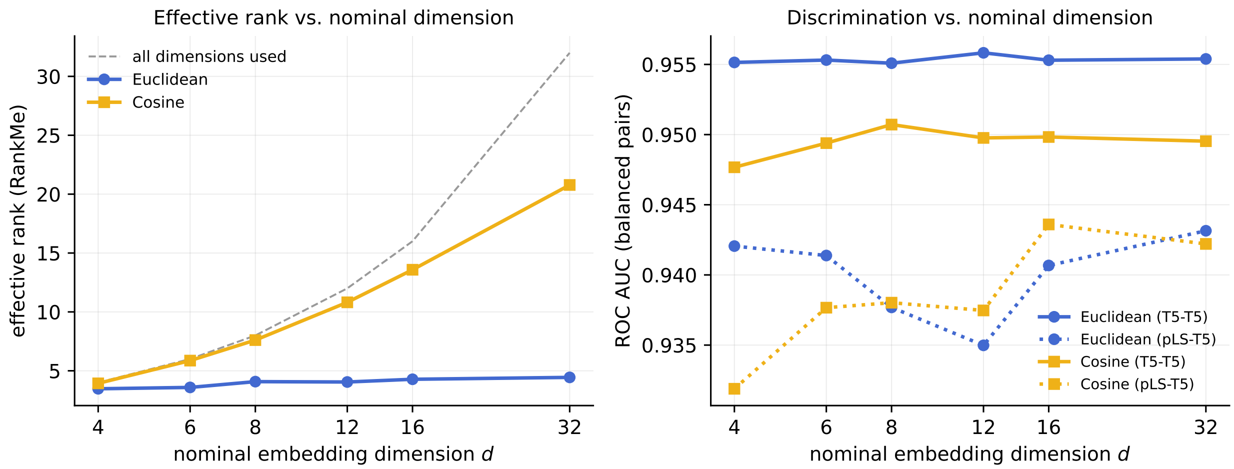

Figure 1: Effective rank and discrimination against nominal embedding dimension. The dashed diagonal is the locus an embedding using all of its dimensions would follow. The Euclidean embedding departs from it immediately and is flat beyond $d \approx 8$, while the cosine embedding continues to track it. Balanced AUC (right) is insensitive to d for both metrics, which is the reported null result that the left panel explains.

to 3.96 and the leading three components still carry 98.2% of the variance. Rank collapse here is driven by *correlation between* dimensions, not by any dimension's marginal variance vanishing, and only a covariance penalty addresses that. This creates a genuine tension with Section 4: a covariance penalty drives $\mathrm{cov} \to I$ and hence $V \to I$, so the very regulariser that would prevent collapse also trivialises the learned precision matrix. One cannot have both.

Scale. A second, related defect: the embedding occupies a scale far below the contrastive margin, so essentially every pair sits inside the margin and the repulsive term never saturates. Setting the margin from the data scale, or removing it entirely in favour of a learned decision scale [14], eliminates this class of error by construction.

4 A global learned metric is a reparameterisation

Write a positive-definite precision matrix as $V = L^\top L$. If the encoder ends in a trainable linear layer with weight W and bias c, then for any pair the bias cancels in the difference and

$$d_V\big(f(a), f(b)\big)^2 = \big(W(h_a - h_b)\big)^\top V \big(W(h_a - h_b)\big) = \big\|LW(h_a - h_b)\big\|^2. \tag{2}$$

Since LW is itself an admissible weight matrix, the model class $\{$encoder ending in W, metric $V\}$ equals $\{$encoder ending in LW, metric $I\}$. A globally learned precision matrix therefore adds no expressive power over the Euclidean metric; it is unidentifiable in every direction except overall scale.

We verify Equation (2) numerically. Planting the exact solution LW reproduces the Mahalanobis distances to a relative error of 1.6×10^{-13}; fitting the same family from scratch reaches a normalised residual of 0.00000, whereas the same family cannot reproduce a per-instance metric (residual 0.358, converged).

When this fails. The equivalence is not universal, and the boundaries matter for anyone applying it:

- A *terminal nonlinearity* breaks it (relative error 0.42), because the metric can no longer be commuted through the final map.
- L_2 *normalisation* breaks it (relative error 0.955), since $Lx/\|Lx\| \neq L(x/\|x\|)$. Metric-learning pipelines that normalise onto the unit hypersphere are therefore unaffected.
- *Weight decay* preserves the distances but not the optimisation problem: $\|W\|_F^2$ and $\|LW\|_F^2$ differ by a factor of 13.3 in our test. The function classes coincide; the regularised problems do not.

The deployed configuration [3] has an unnormalised, linear-output embedding, so the equivalence applies to it.

What does add capacity. The absorption argument fails as soon as the precision depends on the input, because no fixed weight can absorb a pair-dependent transform. The cheapest such model predicts a diagonal log-variance per candidate and scores pairs by the mutual likelihood score [13],

$$s(i,j) = -\tfrac{1}{2}\sum_l \left[\frac{(\mu_i^l - \mu_j^l)^2}{\sigma_i^2(l) + \sigma_j^2(l)} + \log\left(\sigma_i^2(l) + \sigma_j^2(l)\right) \right], \tag{3}$$

whose effective precision $1/(\sigma_i^2 + \sigma_j^2)$ is per-pair and whose log-determinant term has no representation as a distance on the means. This costs one additional linear head.

5 The pairing window has a recall ceiling

The $\Delta R^2 < 0.02$ window is a *blocking* step in the entity-resolution sense [17]: any true duplicate pair whose members fall outside it is never presented to the model and cannot be recovered, whatever the model does. Blocking recall is routinely reported in that literature and bounds every downstream figure, but is not reported for tracking.

Enumerating all true duplicate pairs and asking what fraction falls inside the window, we measure a ceiling of 0.9952 for T5–T5 pairs and 0.9634 for pLS–T5 pairs. The cross-collection ceiling is the tighter of the two, which is consistent with pLS and T5 candidates for the same particle being separated further in η–ϕ than two T5 candidates are.

6 Balanced AUC overstates deployment performance

Pair sampling is balanced 50/50 by construction, whereas the deployed duplicate prior among in-window pairs is a few percent. Because AUC is prior-invariant and integrates over operating points that would never be deployed, it is close to uninformative about deployment.

Table 2 re-evaluates each metric at a realistic prior, at the operating point that preserves 99% of genuine tracks — the quantity that matters, since discarding a non-duplicate destroys a real particle track. A balanced AUC of 0.9566 corresponds to an in-window duplicate recall of 0.189, and 0.188 once the blocking ceiling is folded in.

7 Pairwise decisions are globally inconsistent

"Is the same particle as" is an equivalence relation and must be transitive, but a pairwise scorer has no mechanism enforcing this [18]. We count *closed* triples — those for which all three pairs were actually scored — and ask how many violate transitivity by asserting $a \sim b$ and $b \sim c$ while denying $a \sim c$. Triples with an unscored edge are excluded rather than counted as denials.

Ground-truth labels give 108,465,681 closed triples over 40 densely-paired events and exactly zero violations, confirming the measurement. The trained model is intransitive on 4.0% to 17.5% of closed triples depending on the operating threshold.

A sampling caveat. With the per-event pair cap used in the standard configuration, 98.6% of triples have an unscored edge and the measurement is starved. Generating all in-window pairs instead reduces this to 0.40%, and the figures above use that dense configuration. Anyone repeating this measurement must do the same, or the result will be an artefact of the sampling rather than a property of the scorer.

8 Arbitration is learnable

The embedding answers a symmetric question — are these the same particle? — whereas duplicate removal requires an asymmetric one: which candidate survives? Production answers the second with a fixed collection priority, which carries no information at all for same-collection pairs.

We score a pair with $s(i, j) = g(e_i, e_j) - g(e_j, e_i)$, which is antisymmetric by construction for any weights, so the decision cannot depend on the arbitrary order in which pairs were enumerated. The target is $t5\_pMatched$, the fraction of a candidate's hits belonging to its matched simulated track.

This quantity is quantised to ten values and 54.0% of duplicate pairs are exact ties, which carry no arbitration signal and are excluded from training; the remaining 46.0% are usable. Table 3 reports accuracy by input representation. The head reaches 0.7032 against 0.5 for the priority rule.

The embedding discards what the arbiter needs. Raw features add +0.0576 accuracy over the embedding alone. This is expected rather than surprising: the contrastive objective explicitly minimises $\|e_i - e_j\|$ over duplicates, so in the limit where it succeeds the antisymmetric score is identically zero. We measure duplicate pairs separated to 0.1124 of the embedding scale. The better the embedding is at its own task, the less arbitration information it retains — so an arbiter must read raw features.

9 A threshold with a distribution-free guarantee

Production uses hand-tuned η-binned working points with no statistical guarantee. Since discarding a non-duplicate destroys a real track, the operating question is precisely one of risk control: what threshold removes at most a stated fraction of genuine tracks?

Conformal risk control [19] answers this. For per-event losses $L_i(\lambda)$ that are non-increasing in λ and bounded by B,

$$\hat{\lambda} = \inf \left\{ \lambda : \tfrac{n}{n+1} \hat{R}_n(\lambda) + \tfrac{B}{n+1} \leq \alpha \right\} \tag{4}$$

guarantees $\mathbb{E}[L(\hat{\lambda})] \leq \alpha$ under exchangeability alone.

Table 1: Metric comparison with each metric trained independently under its own objective. RankMe is the effective rank of the 12-dimensional embedding (Section 3); var@3 is the variance captured by the leading three principal components.

Metric	AUC (T5–T5)	AUC (pLS–T5)	RankMe	var@3
Euclidean	0.9566	0.9004	3.96	98.2%
Mahalanobis	0.9297	0.7046	7.47	87.3%
Cosine	0.9566	0.9493	11.05	56.1%

Table 2: Balanced-pair AUC against deployment-relevant performance at a 5% duplicate prior and 99% track efficiency. End-to-end recall folds in the blocking ceiling of the $\Delta R^2 < 0.02$ window.

Metric	Balanced AUC	Recall (in-window)	Ceiling	Recall (end-to-end)
Euclidean	0.9566	0.189	0.9952	0.188
Mahalanobis	0.9297	0.185	0.9952	0.184
Cosine	0.9566	0.189	0.9952	0.188

Two points deserve emphasis. The exchangeable unit is the *event*, not the pair: pairs within an event share a collision and are strongly dependent, and calibrating on pairs inflates the effective sample size by the pairs-per-event factor. In simulation, event-level calibration controls the risk (mean 0.0059 against $\alpha = 0.01$) while pooled-pair calibration violates it (0.0103).

Second, the finite-sample penalty $B/(n+1)$ is paid before any data is seen. With 500 events, half held out for calibration, and the naive bound $B = 1$, it is $1/251 = 0.004$, so $\alpha = 0.005$ has 80% of its budget consumed up front and $\alpha \leq 0.002$ is uncertifiable. Restricting the threshold grid to a range that would plausibly be deployed tightens B and, counter-intuitively, certifies a *more* aggressive threshold, because budget is freed from a worst case that would never occur.

10 Limitations

The equivalence of Section 4 is a known result for linear metric learning [8, 9]; we extend its scope and quantify consequences rather than establishing it. The intransitivity and blocking-ceiling measurements are properties of this pipeline and sample and need not transfer. All results use a single $t\bar{t}$ sample at one pileup. We replicate the central measurements on two disjoint halves of the events, which controls for subset and seed instability but cannot detect a bias shared by the whole sample; external validity across pileup and physics process remains untested. The conformal guarantee requires exchangeability and does not survive a pileup shift, which we demonstrate. We do not report end-to-end tracking efficiency, fake rate, or throughput against the production baseline; those require integration into the CMSSW sequence and are the natural next step. [TODO: integrate and report MTV efficiency/fake/duplicate versus p_T, η, $|d_{xy}|$ at matched efficiency]

11 Conclusion

The deployed learned embedding for LST duplicate removal works, and we do not propose replacing it. We do show that it is operating in a rank-collapsed regime that explains its own reported insensitivity to embedding dimension; that the natural next step of learning a global metric on top of it cannot help, for reasons that are structural rather than empirical; that its pairing window

Table 3: Arbitration accuracy by input representation. The collection-priority rule carries no information for same-collection pairs and is a coin flip by construction. The trained embedding separates duplicate pairs to 0.1124 of the embedding scale, which is why the embedding-only head underperforms.

Input	Accuracy	Accuracy (decisive half)	Ranking AUC
Embedding only	0.6456	0.7619	0.7291
Raw features	0.7032	0.8549	0.8071
Embedding + raw	0.7032	0.8498	0.8073
Collection priority (baseline)	0.5000	0.5000	0.5000

Table 4: Effective rank and discrimination against nominal embedding dimension. Effective rank saturates while the nominal dimension grows by a factor of eight, which is what makes additional dimensions inert.

	Euclidean			Cosine		
d	RankMe	AUC (T5)	AUC (pLS)	RankMe	AUC (T5)	AUC (pLS)
4	3.47	0.9551	0.9421	3.93	0.9477	0.9319
6	3.59	0.9553	0.9414	5.87	0.9494	0.9377
8	4.08	0.9551	0.9377	7.61	0.9507	0.9380
12	4.05	0.9558	0.9350	10.81	0.9498	0.9375
16	4.28	0.9553	0.9407	13.58	0.9498	0.9436
32	4.44	0.9554	0.9431	20.78	0.9495	0.9422

imposes a quantifiable recall ceiling; that its headline evaluation metric substantially overstates deployment performance; and that the one decision it leaves to a fixed heuristic — which duplicate to keep — is learnable from data already present in the sample.

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